RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

*Go, change the world*

**DEPARTMENT OF MATHEMATICS**

**Academic year 2024-2025 (Odd Semester)**

| | | | | | |
|---|---|---|---|---|---|
| **Date** | **02-01-2025** | | **Time** | **10:00 AM-11:30 AM** | |
| **Test** | **Improvement Test Scheme and Solutions** | | **Maximum Marks** | **50** | |
| **Course Title** | **Fundamentals of Linear Algebra, Calculus & Differential Equations** | | | **Course Code** | **MA211TB** |
| **Semester** | **I** | **Programs** | | **AS, CH, IM and ME** | |

| | | |
|---|---|---|
| 1(a) | $(D^2-4D+4)y = x^2e^{3x} \Rightarrow AE \Rightarrow m^2-4m+4=0 \Rightarrow m=2,2 \Rightarrow CF: (c_1+xc_2)e^{2x}$ | 2 |
| | $PI: \frac{1}{D^2-4D+4}\, x^2e^{3x} \Rightarrow e^{3x}\frac{1}{(D+3)^2-4(D+3)+4}\, x^2 \Rightarrow e^{3x}\frac{1}{1+D^2+2D}x^2$ | 1 |
| | $\Rightarrow e^{3x}\,[1+(D^2+2D)]^{-1}\,x^2 \Rightarrow e^{3x}(1+(D^2+2D)-(D^2+2D)^2+\cdots)x^2 \Rightarrow e^{3x}(x^2-4x+6)$ | 2 |
| | $y(x) = CF+PI = (c_1+xc_2)e^{2x}+e^{3x}(x^2-4x+6)$ | 1 |
| 1(b) | $z = 3x^2-4y^3-7x^2y^3\frac{\partial z}{\partial x} = 6x-14xy^3 \Rightarrow \frac{\partial^2 z}{\partial y\partial x} = -42xy^2,\ \frac{\partial z}{\partial y} = -12y^2-21x^2y^2 \Rightarrow \frac{\partial^2 z}{\partial x\partial y} = -42xy^2$ | 2,2 |
| 2 | $m=-2,-2 \Rightarrow CF: (c_1+tc_2)e^{-2t}\ \ PI: \frac{1}{D^2+4D+4}(3\sin t+4\cos t) = 3\,\frac{1}{4D+3}\sin t+4\frac{1}{4D+3}\cos t$ | 2 |
| | $PI: 3\,\frac{4D+3}{16D^2-9}\sin t+4\frac{4D+3}{16D^2-9}\cos t \Rightarrow\ 3\,\frac{4D+3}{-25}\sin t+4\frac{4D+3}{-25}\cos t$ | 2 |
| | $PI:\ -\frac{3}{25}(4\cos t-3\sin t)-\frac{4}{25}(-4\sin t-3\cos t) = \sin t$ | 2 |
| | $x(t) = CF+PI = (c_1+tc_2)e^{-2t}+\sin t,\quad x'(t) = -2e^{-2t}c_1+c_2e^{-2t}-2tc_2e^{-2t}+\cos t$ | |
| | $\Rightarrow x(0) = c_1 \Rightarrow c_1 = 1\ x'(0) = -2c_1+c_2+1 \Rightarrow c_2 = 1$ | 2 |
| | $x(t) = (1+t)e^{-2t}+\sin t$ | 1 |
| 3 | $m = 2i,-2i \Rightarrow CF: c_1\cos 2x+c_2\sin 2x,\ \ W = 2$ | 2,2 |
| | $PI:\ -\cos 2x\ \int\frac{\sin 2x\tan 2x}{2}dx+\sin 2x\ \int\frac{\cos 2x\tan 2x}{2}dx$ | |
| | $= -\cos 2x\ \int\frac{\sec 2x-\cos 2x}{2}dx+\sin 2x\ \int\frac{\sin 2x}{2}dx$ | 2 |
| | $= -\frac{\cos 2x}{4}[\log(\sec 2x+\tan 2x)-\sin 2x]-\frac{\sin 2x}{2}\cos 2x = -\frac{\cos 2x}{4}[\log(\sec 2x+\tan 2x)]$ | 3 |
| | $y = CF+PI = c_1\cos 2x+c_2\sin 2x-\frac{\cos 2x}{4}[\log(\sec 2x+\tan 2x)]$ | 1 |
| 4 | $let\ x = e^t \Rightarrow t = \log x,\ xy' = Dy, x^2y'' = D(D-1)y, D = \frac{d}{dx}\ (D^2-4D+5)y = e^{2t}\sin t$ | 2 |
| | $m = 2-i, 2+i \Rightarrow CF: e^{2t}(c_1\cos t+c_2\sin t);\ PI:\ \frac{1}{D^2-4D+5}\ e^{2t}\sin t = e^{2t}\frac{1}{D^2+1}\sin t$ | 2,2 |
| | $\Rightarrow e^{2t}t\frac{1}{2D}\ \sin t = -\frac{e^{2t}t\cos t}{2} \Rightarrow y(x) = x^2[c_1\cos(\log x)+c_2\sin(\log x)]-\frac{x^2\log x\cos(\log x)}{2}$ | 2,2 |
| 5(a) | $V = e^{3x+4y}\cos(5z)\ V_x = 3e^{3x+4y}\cos(5z), V_{xx} = 9e^{3x+4y}\cos(5z),$ | 2 |
| | $V_y = 4e^{3x+4y}\cos(5z), V_{yy} = 16e^{3x+4y}\cos(5z),\ \ V_z = -5\,e^{3x+4y}\sin(5z), V_{zz} = -25\,e^{3x+4y}\cos(5z)$ | 2,2 |
| | $V_{xx}+V_{yy}+V_{zz} = 0$ | 1 |
| 5(b) | $u_x = u_r\frac{1}{y}+u_s\frac{1}{z}+u_t\left(-\frac{z}{x^2}\right) \Rightarrow xu_x = u_r\frac{x}{y}+u_s\left(\frac{x}{z}\right)-u_s\left(\frac{z}{x}\right);\ \ y\,u_y = -u_r\left(\frac{x}{y}\right)+u_s\left(\frac{y}{z}\right)$ | 2 |
| | $zu_z = -u_s\left(\frac{y}{z}\right)+\ u_t\left(\frac{z}{x}\right);\ \ xu_x+yu_y+zu_z = 0$ | 1,2 |

*****